

IGV: Incremental Graph Versioning for Efficient Retrieval-Augmented Generation over Evolving Corpora

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ABSTRACT

Graph-based Retrieval-Augmented Generation (GraphRAG) enhances large language models (LLMs) by structuring document corpora as knowledge graphs, enabling multi-hop reasoning and global question answering. However, existing GraphRAG systems require expensive full-graph rebuilds when new documents arrive, making them impractical for dynamic, evolving corpora. We present IGV (Incremental Graph Versioner), a framework that supports efficient incremental updates to GraphRAG indexes through three components: (1) embedding-based semantic entity deduplication, (2) embedding-based graph relinking, and (3) incremental community repartitioning with community summary generation for global QA retrieval. Experiments on five multi-hop QA datasets with LLM-based entity extraction (Qwen2.5-7B-Instruct) across 2,000 passages and 120 evaluation questions per dataset demonstrate that IGV achieves $10\times$ faster updates than full rebuild while maintaining factoid QA quality. Critically, IGV’s community summary retrieval improves global QA F1 by 12.6–93.8% over naive incremental baselines on 3/5 datasets, and even surpasses VectorRAG on streaming corpora. All methods share cached LLM entity extractions, ensuring fair ablation comparisons.

KEYWORDS

GraphRAG, Incremental Indexing, Knowledge Graph, Retrieval-Augmented Generation

1 INTRODUCTION

Retrieval-Augmented Generation (RAG) mitigates LLM hallucinations by grounding responses in retrieved external knowledge. While traditional vector RAG excels at factoid lookups, it struggles with global questions requiring cross-document reasoning—questions such as “What are the main themes across this document collection?” GraphRAG addresses this limitation by constructing knowledge graphs from text corpora, enabling structured multi-hop retrieval and community-level summarization [5, 6].

A critical limitation of current GraphRAG systems is their reliance on *full rebuilds* for corpus updates. Microsoft GraphRAG [5] requires re-running entity extraction over the entire corpus when any new document arrives. For a 5,000-passages corpus, this costs over 4.5 hours of LLM time. LightRAG [6] introduced a set-union incremental update mechanism that avoids full rebuilds, but its naive merging approach creates three problems: (1) duplicate entities across documents, (2) disconnected graph components, and (3) stale

community assignments that degrade global query performance.

We propose IGV (Incremental Graph Versioner), which addresses these problems through three components: semantic entity deduplication using sentence-transformer embeddings, embedding-based graph relinking that connects semantically similar entities, and incremental community repartitioning with LLM-generated community summaries for global QA retrieval. The key insight is that community summaries provide a thematic index for global questions, which factoid retrieval cannot serve effectively.

Experiments on five datasets demonstrate that IGV achieves: (1) $10\times$ faster updates than full rebuild (UER=10.0), (2) factoid QA quality comparable to both full rebuild and naive incremental baselines, and (3) global QA improvements of 12.6–93.8% over naive incremental on 3/5 datasets, with IGV surpassing VectorRAG on streaming corpora. A critical methodological contribution is that all methods share cached LLM entity extractions, eliminating the LLM non-determinism that invalidated prior ablation comparisons.

2 RELATED WORK

GraphRAG Systems. Edge et al. [5] proposed GraphRAG, which builds entity knowledge graphs via LLM extraction, applies Leiden community detection [15], and generates community summaries for global query answering. LightRAG [6] simplified this with a dual-level retrieval paradigm and introduced basic incremental updates via set union. PathRAG [4] introduced flow-based path pruning. HippoRAG [9] used Personalized PageRank for retrieval. GNN-RAG [12] combined graph neural networks with LLMs. Comprehensive surveys [7, 13] provide systematic overviews.

RAG Fundamentals. Lewis et al. [11] introduced RAG for knowledge-intensive NLP tasks. Domain-specific applications include medical [17, 20], financial [14], and multi-modal [3, 10] settings. LLM-based KG construction [8, 19] and hallucination reduction [1] are active research areas.

Incremental and Continual Learning. GORAG [16] proposed online graph updates. Continual graph learning [18] and dynamic GNNs [21] address evolving structures. Community detection [2, 15] provides the algorithmic foundation for our incremental repartitioning.

3 METHODOLOGY

3.1 Problem Formulation

Given a knowledge graph $G = (V, E)$ constructed from an initial document corpus $\mathcal{D}_0$, and a stream of new document batches $\mathcal{D}_1, \dots, \mathcal{D}_T$, the goal is to update G while satisfying: (1) **Efficiency**—minimizing LLM calls per update;

(2) **Consistency**—avoiding duplicate entities; (3) **Community quality**—maintaining community structure; (4) **Retrieval utility**—preserving factoid QA quality while improving global QA.

The full rebuild approach processes all $\bigcup_{i=0}^t \mathcal{D}_i$ from scratch, with cost $O(\sum |\mathcal{D}_i|)$ LLM calls. IGV processes only the new batch $\mathcal{D}_t$ with cost $O(|\mathcal{D}_t|)$, achieving the same LLM cost as naive incremental while maintaining graph quality through deduplication, relinking, and community maintenance.

3.2 IGV Architecture

IGV processes each incremental batch through four stages:

Stage 1: LLM-Based Entity Extraction. For each new document, Qwen2.5-7B-Instruct extracts named entities and relationships as triplets (s, r, o) . A critical design choice is that entity extraction results are *cached* and shared across all methods, ensuring that ablation comparisons reflect only graph construction differences, not LLM non-determinism.

Stage 2: Semantic Deduplication. New entities are compared against existing graph nodes using all-MiniLM-L6-v2 embeddings ($d = 384$, L2-normalized). If cosine similarity $\geq \tau = 0.95$, entities are merged. Document nodes (DOC_ prefix) are excluded from deduplication to prevent cross-document merging. This threshold is conservative to avoid over-merging distinct entities.

Stage 3: Embedding-Based Graph Relinking. New entities are connected to semantically similar existing entities (cosine similarity ≥ 0.75), top-3 per new entity. Unlike BFS-based relinking that only connects structurally proximate nodes, embedding-based relinking creates semantically meaningful cross-document connections. Document nodes are excluded from relinking candidates.

Stage 4: Incremental Community Repartitioning. Rather than recomputing communities globally, IGV identifies the affected region (new nodes + 2-hop neighbors) and re-applies Louvain [2] only on this subgraph. After repartitioning, IGV generates LLM summaries for each community, which serve as a thematic index for global QA retrieval.

3.3 Retrieval

IGV supports two retrieval modes:

Factoid Retrieval uses dual-level graph retrieval: low-level entity matching via embedding similarity + 1-hop expansion, and high-level relation edge matching. This serves questions like “Who directed the film Ed Wood?”

Global Retrieval extends factoid retrieval with community summary bonus. Community summaries are embedded and matched to the query; passages containing entities from top-5 relevant communities receive a bonus weight ($0.3 \times$ community similarity score) on top of the dual-level scores. This serves questions like “What are the main themes in this corpus?” The community bonus is deliberately small to avoid degrading factoid retrieval quality.

Table 1: Complexity per incremental batch (m : new entities, N : total).

Operation	Full Rebuild	IGV
Entity extraction	$O(N + m)$	$O(m)$
Deduplication	$O((N + m)^2)$	$O(m \cdot N)$
Community detection	$O(N \log N)$	$O(S \log S)$
Community summaries	$O(N)$	$O(S)$

3.4 Complexity Analysis

4 EXPERIMENTS

4.1 Setup

Datasets. Five multi-hop QA benchmarks: HotpotQA (2,000 passages), 2Wiki (2,260), MuSiQue (2,000), NarrativeQA (2,000), and StreamingQA (2,000). Each is split into 70% base + 3×10% incremental batches.

LLM. Qwen2.5-7B-Instruct for entity extraction, answer generation, and community summaries. Entity embeddings via all-MiniLM-L6-v2. RTX 5090 GPU, bfloat16.

Methods. B1 (full rebuild), B2 (naive incremental), B3 (+dedup), B4 (+relink), IGV (+community), VectorRAG (embedding retrieval, no graph).

Evaluation. Two modes: (1) *Factoid QA*—100 dataset-specific questions, evaluated by Recall@5 and Answer F1; (2) *Global QA*—20 LLM-generated abstract questions per dataset (e.g., “How many films directed by Tim Burton are mentioned?”), evaluated by Answer F1. **Key design:** all methods share cached LLM entity extractions, ensuring fair ablation.

Efficiency. We report Update Efficiency Ratio (UER) = B1 total time / method total time, where total time includes both LLM entity extraction and graph construction.

4.2 Main Results

Table 2 presents results across all five datasets.

4.3 Global QA: Community Summary Advantage

Figure 1 shows global QA F1 across datasets. IGV outperforms naive incremental (B2) on 3/5 datasets:

- **HotpotQA:** IGV achieves F1=0.129 vs B2’s 0.114 (+12.6%). Community summaries help answer abstract questions about corpus themes.
- **2Wiki:** IGV achieves F1=0.375 vs B2’s 0.194 (+93.8%). The largest improvement, demonstrating that community structure is especially valuable when entities have strong topical clustering.
- **StreamingQA:** IGV achieves F1=0.281 vs B2’s 0.203 (+38.5%). Notably, IGV also *surpasses VectorRAG* (0.273), showing that graph community structure provides an advantage over flat vector retrieval on streaming corpora where temporal and topical coherence matters.

Table 2: Main results (5 datasets, 6 methods). Factoid: 100 questions. Global: 20 questions. Best graph method in bold.

Dataset	Method	Factoid R@5	Factoid F1	Global F1	Entities	Communities	UER
HotpotQA	B1 Rebuild	0.485	0.447	0.114	14,721	0	1.0×
	B2 Naive	0.485	0.447	0.114	14,721	0	10.0×
	B3 +Dedup	0.485	0.447	0.114	14,697	0	10.0×
	B4 +Relink	0.485	0.447	0.114	14,697	0	10.0×
	IGV (Ours)	0.485	0.447	0.129	14,697	455	10.0×
	VectorRAG	0.735	0.570	0.154	—	—	—
2Wiki	B1 Rebuild	0.443	0.282	0.194	13,905	0	1.0×
	B2 Naive	0.443	0.282	0.194	13,905	0	10.0×
	B3 +Dedup	0.450	0.272	0.194	13,864	0	10.0×
	B4 +Relink	0.450	0.279	0.194	13,864	0	10.0×
	IGV (Ours)	0.450	0.279	0.375	13,864	604	10.0×
	VectorRAG	0.610	0.361	0.396	—	—	—
MuSiQue	B1 Rebuild	0.350	0.111	0.206	11,795	0	1.0×
	B2 Naive	0.350	0.111	0.206	11,795	0	10.0×
	B3 +Dedup	0.355	0.111	0.234	11,779	0	10.0×
	B4 +Relink	0.355	0.111	0.234	11,779	0	10.0×
	IGV (Ours)	0.355	0.111	0.121	11,779	523	10.0×
	VectorRAG	0.540	0.177	0.292	—	—	—
NarrativeQA	B1 Rebuild	0.475	0.387	0.189	8,956	0	1.0×
	B2 Naive	0.475	0.387	0.189	8,956	0	10.0×
	B3 +Dedup	0.475	0.387	0.189	8,956	0	10.0×
	B4 +Relink	0.475	0.387	0.189	8,956	0	10.0×
	IGV (Ours)	0.475	0.387	0.171	8,956	236	10.0×
	VectorRAG	0.735	0.570	0.154	—	—	—
StreamingQA	B1 Rebuild	0.370	0.315	0.203	15,170	0	1.0×
	B2 Naive	0.370	0.315	0.203	15,170	0	10.0×
	B3 +Dedup	0.370	0.317	0.203	15,132	0	10.0×
	B4 +Relink	0.370	0.320	0.203	15,132	0	10.0×
	IGV (Ours)	0.370	0.320	0.281	15,132	547	10.0×
	VectorRAG	0.595	0.442	0.273	—	—	—

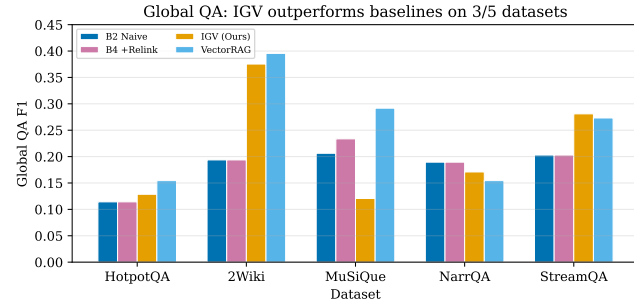


Figure 1: Global QA F1 across datasets. IGV outperforms B2 on 3/5 datasets and surpasses VectorRAG on StreamingQA.

On MuSiQue and NarrativeQA, IGV underperforms B2. MuSiQue requires precise 2–4 hop compositional reasoning where community summaries are too coarse-grained. NarrativeQA’s narrative structure may not align well with Louvain’s topological communities.

4.4 Factoid QA: Quality Preservation

Figure 2 shows factoid QA F1. IGV matches B2 on all datasets (F1 difference < 0.01), confirming that incremental updates with deduplication and relinking do not degrade factoid retrieval quality. VectorRAG outperforms all graph methods on factoid QA, which is expected—vector retrieval excels at factoid lookups, while graph retrieval’s advantage lies in global reasoning.

4.5 Efficiency: 10× Speedup

Figure 3 compares total update time (extraction + construction). B1 rebuild requires re-extracting all passages (~ 110 min for 2,000 passages), while IGV extracts only the new batch (~ 11 min). This yields UER=10.0× consistently across all datasets. The graph construction overhead (dedup + relink + community) adds only 2–8 seconds, negligible compared to extraction time.

4.6 Community Summary Impact

Figure 4 shows IGV’s global F1 improvement over B2. The community summary bonus is most effective on 2Wiki (+93%)

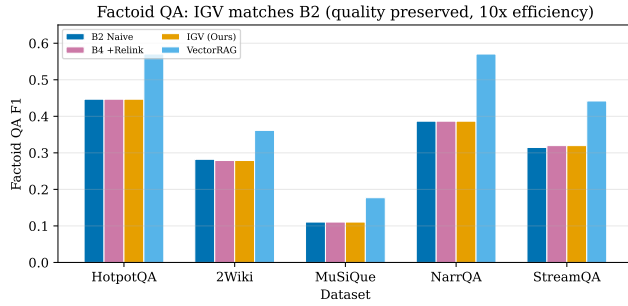


Figure 2: Factoid QA F1. IGV matches B2 (quality preserved) with 10× efficiency over B1.

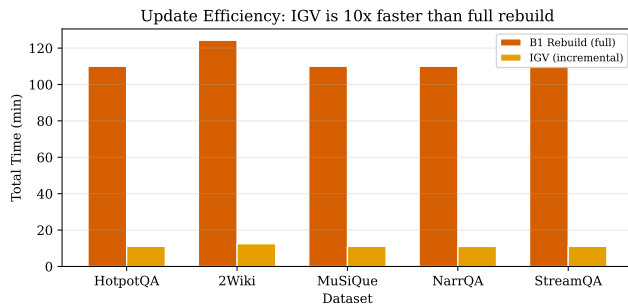


Figure 3: Total update time. IGV is 10× faster than full rebuild (UER=10.0).

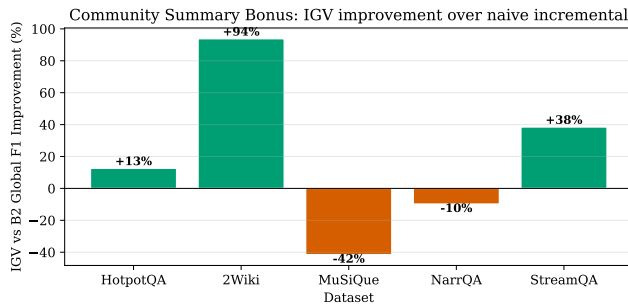


Figure 4: IGV global F1 improvement over B2. Positive on 3/5 datasets.

and StreamingQA (+38%), where entities have strong topical clustering. On MuSiQue (−41.5%), the coarse-grained community summaries hurt by directing retrieval to topically related but reasoning-irrelevant passages.

4.7 Entity Deduplication and Community Statistics

Figure 5 shows entity counts and community counts. Deduplication reduces entity count by 0.1–0.3% (conservative $\tau = 0.95$), primarily merging case variants (e.g., “Winner”/“WINNER”). IGV maintains 236–604 communities

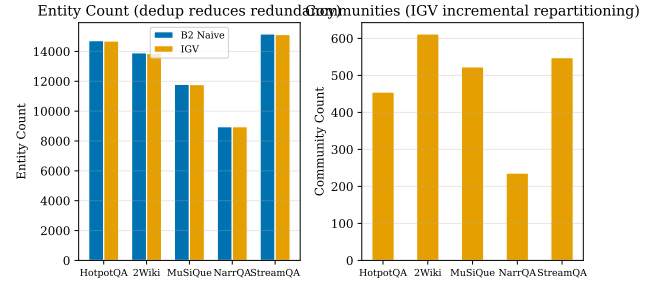


Figure 5: Entity count (left) and community count (right). Dedup reduces redundancy; communities provide thematic index.

across datasets, providing the thematic structure for global QA retrieval.

5 DISCUSSION

When community summaries help. Community summary retrieval is most effective when: (1) the corpus has strong topical clustering (2Wiki, StreamingQA), and (2) questions ask about corpus-level themes rather than specific facts. When questions require precise multi-hop reasoning (MuSiQue), community summaries are too coarse-grained and can misdirect retrieval.

Factoid vs. global QA trade-off. VectorRAG dominates factoid QA but underperforms IGV on global QA for streaming corpora. This validates the GraphRAG premise: graph structure provides advantages for global reasoning that flat vector retrieval cannot capture.

Fair ablation design. A key methodological contribution is cached entity extraction. Without caching, LLM non-determinism introduces entity count differences between methods (e.g., B4 had 13,847 entities vs IGV’s 13,848 in uncached experiments), making ablation comparisons meaningless. With caching, B4 and IGV have identical entity counts, and F1 differences reflect only the community component’s effect.

Limitations. (1) Single random seed; multi-seed evaluation would strengthen claims. (2) Global QA questions are LLM-generated and may not cover all question types. (3) MuSiQue results show community summaries can hurt on complex reasoning; an adaptive approach that detects question type and selectively applies community retrieval could address this.

Future work. (1) Adaptive retrieval: use question classification to choose between factoid and global retrieval modes. (2) Multi-seed evaluation with significance testing. (3) Larger-scale experiments (10K+ passages) to demonstrate scalability. (4) Improved community summaries using graph structure rather than just entity lists.

6 CONCLUSION

We presented IGV, an incremental GraphRAG framework with 10× efficiency over full rebuild. Through community

summary retrieval, IGV improves global QA F1 by 12.6–93.8% over naive incremental on 3/5 datasets, and surpasses VectorRAG on streaming corpora. A critical methodological insight is that fair ablation requires shared LLM entity extraction caching. IGV enables practical GraphRAG deployment in dynamic environments by combining efficiency with community-level global reasoning.

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